Auction Summarization with Large Language Models: A Cross-Model Comparison

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Abstract

This is an empirical study on the 'Auctions with LLM Summaries' written by Google Researchers - Avinava Dubey, Zhe Feng, Rahul Kidambi, Aranyak Mehta, Di Wang. This paper studied the allocation mechanisms in ad auctioning and proposed a study that uses GPA and a LLM module working together to provide a welfare maximizing summary in an incentive compatible manner. We use the proposed mechanism from the paper and compare the welfare results generated by three different LLMs in our study.

1 Introduction

This paper explores how ad auctions should work when AI writes the final results. Unlike traditional GSP or VCG auctions that assign fixed slots (Edelman et al., 2007; Ausubel et al., 2006), prominence-based auctions assign each ad a continuous attention score in the LLM-generated summary (Dubey et al., 2024). The core challenge is that the LLM acts as a black box, the auction cannot see the final output before making decisions. We focus on generalized proportional allocation (GPA) with a LLM module as the mechanism for welfare-maximizing, incentive-compatible ad placement.

1.1 Our contribution

We use the foundations of the original paper for our study. The difference is that the original paper uses only Gemini Pro to evaluate the prominence-based auction design embedded in a factorized framework. We take the same approach but evaluate this auction design in three LLMs: Gemini 2.5 Flash, Claude Haiku 4.5, and ChatGPT 4o Mini.

Since the original code was not available, we reproduced the methods based on the paper's description. The project was split evenly between us. Each of us ran 250 queries, for a total of 500 queries per LLM. We also divided the work for the presentation and paper equally.

2 Key Concepts

2.1 Welfare Maximization and Incentive Compatibility

Welfare Maximization is the total expected value created by an allocation summed across all advertisers. The auction maximizes:

$$\text{welfare} = \sum_i \text{bid}_i \times \text{base CTR}_i \times \mathrm{ROUGE}_{\beta,i} \times \text{position}_i \tag{1}$$

where bid_i is the advertiser's payment per click, base CTR_i is the ad's expected click-through rate, $\mathrm{ROUGE}_{\beta,i}$ measures summary quality after compression, and position_i discounts ads appearing lower in the paragraph.

The auction is incentive compatible, meaning each advertiser's best strategy is to report their true value per click. This is enforced by **Myerson's Lemma**, which guarantees that a monotonic allocation rule paired with a specific payment formula makes the auction cheat-proof. Monotonicity ensures that greater prominence, such as, more words, stronger mentions, or better placement can yield more user attention. In other words, no advertiser can benefit by misreporting their value.

2.2 eCPM and CTR

To evaluate an ad's value relative to its likelihood of being clicked, we use eCPM (effective cost per mille):

$$\text{eCPM} = \text{CTR} \times \text{CPC Bid} \times 1000$$

where CTR is the click-through rate, CPC Bid is the advertiser's willingness to pay per click, and the factor of 1000 converts expected revenue per impression to per 1,000 impressions.

An ad's CTR therefore depends on its own quality and relevance, the prominence the auction assigns it, and how the LLM represents it in the final summary. The predicted CTR (pCTR) module estimates:

$$\text{pCTR}_i(\text{Prom}, z)$$

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where Prom is the prominence allocation, z represents ad features such as the landing page, and i denotes a specific advertiser.

2.3 Defining α and β

β is the compression quality penalty exponent, while α is the sharpness parameter of the GPA allocation rule, derived as $\alpha = \frac{1}{1-\beta}$. Together, these parameters determine how prominence translates into word allocation and expected click-through rate.

3 Methodology

3.1 3 Mechanisms Used

3.1.1 GPA + LLM

First, the auction module assigns each ad a word share proportional to its value, measured by the eCPM. Then, the LLM module compresses each ad to its word share according to the budget set. The tradeoff is the mechanism has a ROUGE Score < 1. Meaning some content quality is lost in compression and at larger word budgets this compression cost outweighs the benefit of proportional allocation.

3.1.2 Greedy

This can be treated as the baseline model. This mechanism doesn't use a LLM. It ranks all ads by eCPM and shows them in full until the word budget runs out. It doesn't fare well with tight budgets as it can fit less ads in it. It performs the worst when the space is scarce, but performs well when the budget is high and eventually takes over GPA + LLM.

3.1.3 POS-FL - Position Auction With Fixed Length

This mechanism has no optimization, it is the perfect middle ground. It uses the LLM to compress ads but doesn't allocate words proportionally. It gives every ad an equal share of the total word budget.

3.2 Variables Used

Ads are LLM-generated using Gemini. There are 2–4 ads per query, ≤ 40 words each, representing different advertisers bidding for placement. The first 500 queries out of a 1000 generated queries are used.

According to the eCPM and Welfare Formula:

- Bids are drawn from LogNormal(0.5, 1) - so they vary widely (0.1 to 10+).
- Base CTRs are drawn from Uniform[0,1] - an estimate of how likely the ad gets clicked.
- $\beta - 1/4, 1/3, 1/2$ - used to calculate α
- number of words - 30,40,50,60,70,80

4 Results

4.1 Finding 1 - GPA + LLM Wins Overall

We compare the welfare values across each LLM for the three mechanisms mentioned above, at different values of Beta: $\beta - 1/4, 1/3, 1/2$.

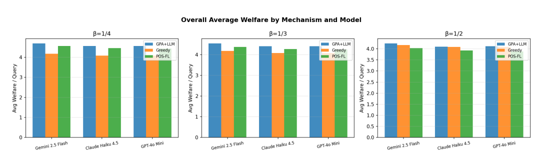

Figure 1: Comparing mechanisms by models according to β values

Looking at the welfare values averaged across all word budgets, we can see that GPA + LLM works the best.

4.2 Finding 2 - GPA + LLM has Advantage in Tight Budgets

When the word budget is as low as 30 words, the Greedy Mechanism wastes it on one full ad (or may not even fit one) and then runs out of room to include more ads. GPA + LLM compresses each ad proportionally by prominence, fitting the top ads within the budget. It has approximately 40% higher welfare than Greedy at β = 1/4. However, as the budget grows to say 60 or 70 words, Greedy fits full ads without compression, yielding a ROUGE score close to 1 and making the strategy of showing the best ads in full text the winner. This is discussed in the next subsection.

4.3 Finding 3 - Greedy Overtakes at Large Budgets

A smaller β allows the LLM to produce shorter summaries with minimal welfare penalty. A higher β increases the cost of each word cut, meaning compressed summaries must be of higher quality to compete with showing the full ad. The crossover budget, at which Greedy begins to outperform GPA + LLM, occurs at approximately 70–80 words for $\beta = 1/4$, 70 words for $\beta = 1/3$, and 60 words for $\beta = 1/2$.

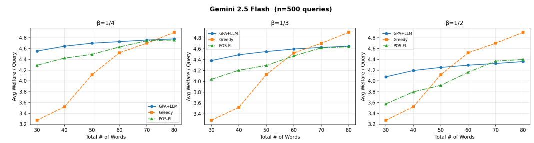

Figure 2: Welfare vs Word Budget: gemini-2-5-flash

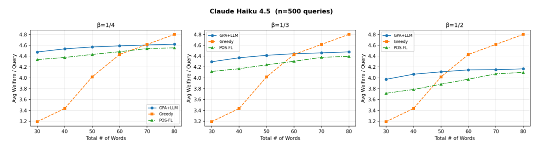

Figure 3: Welfare vs Word Budget: claude-haiku-4-5-2025100

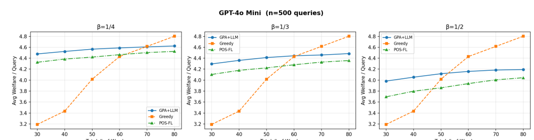

Figure 4: Welfare vs Word Budget: GPT-4o-mini

At a full glance, we can see how the gap between GPA + LLM and Greedy looks for different β values.

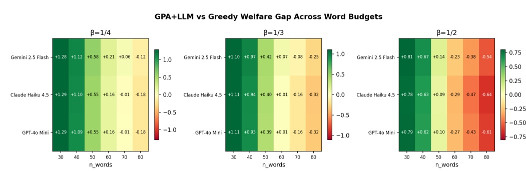

Figure 5: Welfare Gap across word budgets and models at different β values

Green indicates regions where GPA+LLM outperforms Greedy, while red indicates the reverse. Moving left to right, the gap between the two mechanisms narrows. The red region appears earlier as β increases, reflecting the lower crossover budget at higher penalty values.

4.4 Finding 4 - Model Comparison

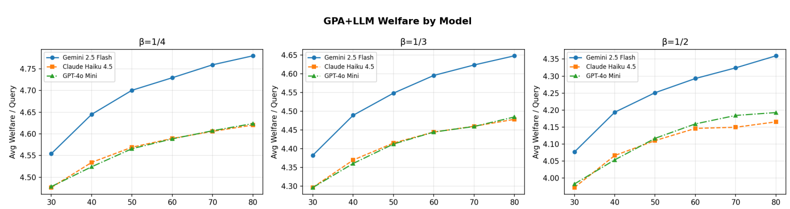

Figure 6: Comparing different LLMs for GPA + LLM for different β values

Gemini achieves the highest welfare, likely due to a home-field advantage as the queries were generated using Gemini. Claude and GPT perform comparably across most settings. However, Claude exhibits a notable welfare dip at β = 1/2 under larger word budgets.

5 Conclusion

In conclusion, GPA + LLM mechanism wins overall, the biggest difference is when space is tight in the LLM summary. Using this mechanism does not differ in the LLM used. Overall, it is not the LLM that matters it is the mechanism used to drive the ad auction that matters.

6 References

Ausubel, L.M. & Milgrom, P. (2006) — "The Lovely but Lonely Vickrey Auction," in Combinatorial Auctions (Cramton, Shoham & Leyton-Brown, eds.), MIT Press, pp. 17–40.

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